CS3235 WWLLMS · SECTION D · CIE3 LAB ASSESSMENT

Multilingual Lecture-to-Notes & Quiz Generator

WWLLM Lecture Studio: Project 5 of the AI4Bharat and BharatGen mini-projects



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github.com/Tamishsri/WWLLM-CIE-3

1. The problem statement

The CIE3 lab assessment asks each student to build one of five projects on India's sovereign models and datasets from AI4Bharat and BharatGen. The objective is a usable, end-to-end Indian-language AI application built on pretrained models, with a suitable frontend and measurable evaluation. This submission is Project 5.

> ### 5. Multilingual Lecture-to-Notes & Quiz Generator
>
> | | |
> |---|---|
> | Application | Transcribe a recorded lecture, create clear notes and five MCQs, and optionally translate the notes. |
> | Models / data | AI4Bharat IndicConformer or BharatGen speech model; Param1; IndicTrans2 |
> | Concepts | ASR, summarization, quiz generation |
> | Evaluation | WER, factual consistency, MCQ quality |
> | All projects | 30 to 50 representative test cases, a failure analysis, a usable frontend and a GitHub repository |
>
> CS3235 Mini-Project Ideas

A recording is slow to revise from and can't be searched. In India it may also mix languages, or be given in a language the student reads less easily. This project turns one recording into material a student can read, test themselves on and return to in their own language.

2. What the brief asked for and what was delivered

The brief asks for	Delivered	Status
Transcribe a recorded lecture	Silero-VAD and Whisper on uploaded audio or video	Met
Clear notes	Structured English notes from Param-1, streamed as they are written	Met
Five MCQs	Five questions from Param-1 by default, adjustable from 3 to 10	Met, quality mixed
Optional translation	Notes and quiz in ten Indian languages	Met
IndicConformer or a BharatGen speech model	Whisper. The IndicConformer code path is in place but unrun, because the model is gated	Substituted
Param1	BharatGen Param-1 2.9B Instruct	Met
IndicTrans2	NLLB-200. The IndicTrans2 code path is in place but unrun, because the model is gated	Substituted
WER	Measured on 40 lectures: average 0.258	Met
Factual consistency	Hand check of the sample notes: 16 of 17 statements correct	One lecture only
MCQ quality	Hand check of the sample quiz: 1 of 5 questions fully sound	One lecture only
30 to 50 test cases	40 lectures with reference transcripts	Met
Failure analysis	Section 7, and `outputs/evaluations/failure_analysis.csv`	Met
Usable frontend	A custom web studio, plus a Streamlit app	Met
GitHub repository	github.com/Tamishsri/WWLLM-CIE-3	Met

Assessment rubric	Where it is covered
Problem definition & relevance	Section 1
Model / dataset selection	Section 4
Working end-to-end application	Sections 3 and 5
GenAI / RAG / agentic integration	Param-1 generates the notes and quiz (Section 4). There is no RAG or agent
Quantitative testing & error analysis	Sections 6 to 9
Demo, documentation & repository	Sections 5 and 11, and the repository README

3. What was made

WWLLM Lecture Studio is a web application. Upload a lecture, or open the built-in sample, and it produces a timed transcript, structured notes, a practice quiz and translations, each stage streaming its progress to the browser. It runs on an ordinary laptop with no GPU.

40
test lectures in four spoken languages

0.258
average word error rate

10
translation languages

122
automated tests passing

Beyond the brief, the application adds these:

- Every output links back to the audio. A note section, a quiz answer or a transcript word is one click from the moment it was spoken.
- Search works across scripts by sound, so the English word "gradient" finds the same word in Hindi or Tamil text.
- A concept map places the ideas in a lecture along its timeline.
- Theatre mode plays the lecture with word-by-word captions and a rangoli pattern drawn from the loudness of the speech.
- The Benchmark page scores any set of predictions, shows a word-by-word comparison and exports failures as CSV.
- Results are stored by content, so a lecture is only processed once, and a badge on each result says what produced it.

4. How it works

Audio or video Upload, or the sample › **Silero-VAD** Finds each stretch of speech › **Whisper** Speech to text › **Param-1** Notes and quiz › **NLLB-200** Ten Indian languages

Each stage tries three things in order: a stored result for this exact input, the real model, then an offline method. The studio is a Python backend on Starlette that streams events to a browser frontend written in plain JavaScript modules.

Stage	Model	Job	Time on the sample
Speech detection	Silero-VAD v5	Cuts the audio into 26 stretches of speech with timestamps	7 s
Transcription	whisper-hindi-small	Hindi speech to text, one stretch at a time	2.0 min
Notes	Param-1 2.9B Instruct	Structured English notes	9.6 min
Quiz	Param-1 2.9B Instruct	Five multiple-choice questions	8.2 min
Translation	NLLB-200	Notes and quiz into one language, keeping the answer key	64 s

Measured on the 98-second sample, on an Intel i5-1235U with 16 GB of RAM and no GPU. All stages together take about 31 minutes.

Model selection

The brief says to confirm checkpoint availability, licensing, language coverage, GPU requirements and suitability before using a model. Those checks changed two choices. IndicConformer and IndicTrans2 are gated on Hugging Face and need a personal login, and IndicConformer also needs the NeMo toolkit. The open alternatives run on a CPU with no login: Whisper, with models fine-tuned for Hindi, Tamil and Kannada, and NLLB-200, which uses the same language codes as IndicTrans2. Param-1 is used as the brief asks. The code tries the AI4Bharat models first when their toolkits are installed.

Search across scripts

Search needs no model. Hindi, Tamil and the other major Indic scripts share one Unicode layout, so a single table reduces them, and Latin spelling, to the same consonant skeleton:

```
gradient  →  K R T N T      ग्रेडिएंट  →  K R T N T      கிரேடியன்ட்  →  K R T N T
```

5. The application in use

Every screenshot comes from the running application and shows real model output.

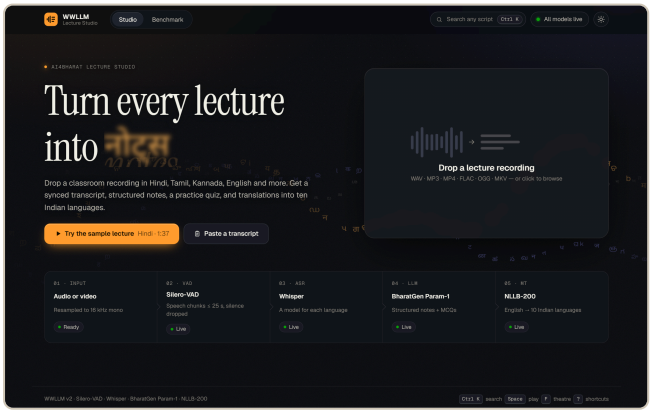


Figure 1. The home page. Upload a recording, paste a transcript, or open the sample lecture.

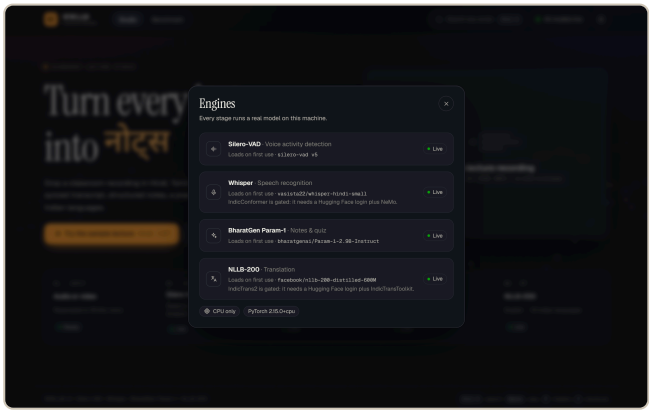


Figure 2. The engines panel shows the model behind each stage and whether it can run on this machine.

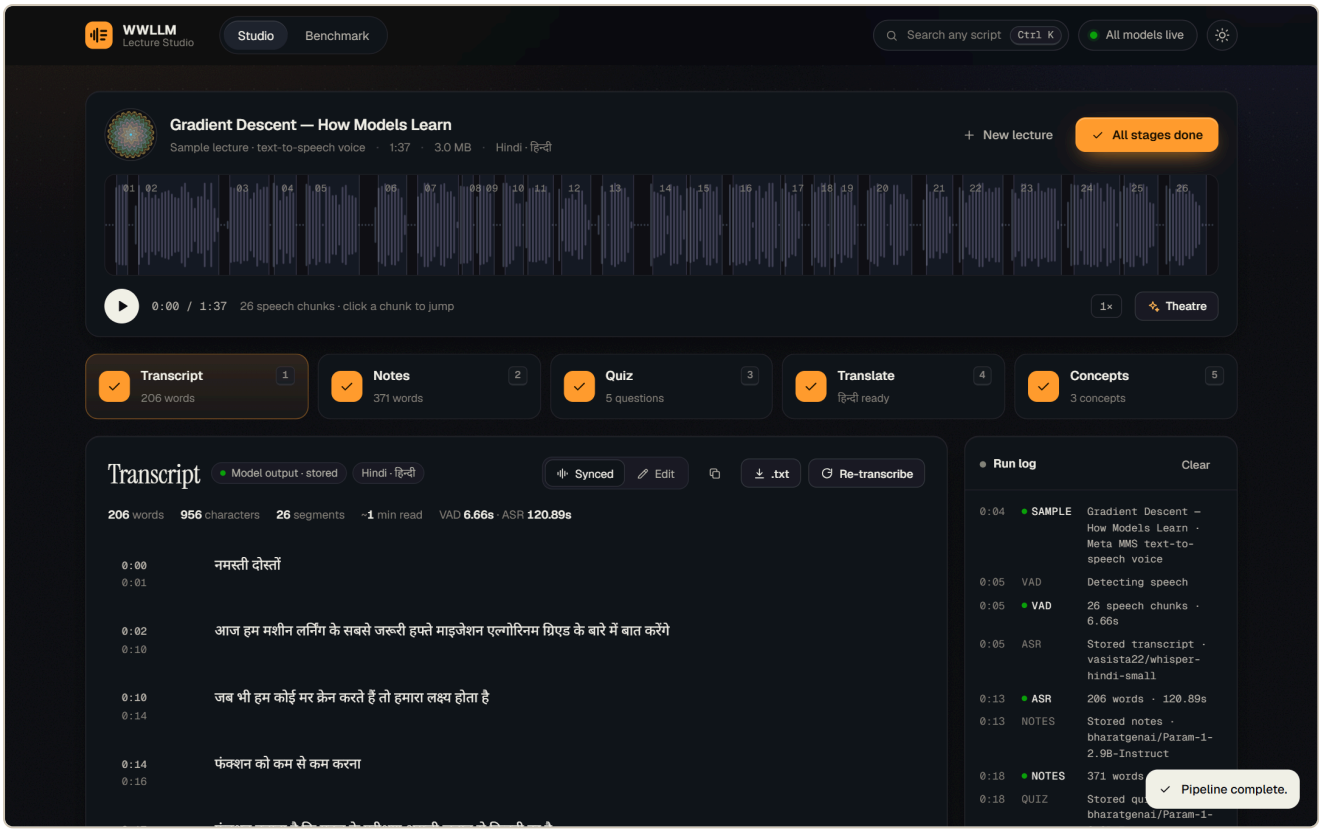


Figure 3. The transcript from Whisper, 206 words of Hindi across 26 stretches of speech. Words light up during playback, and clicking one jumps to that moment.



Figure 4. Notes from Param-1, which read the Hindi transcript and wrote 371 words of English. Each section has a chip that plays the matching part of the lecture.

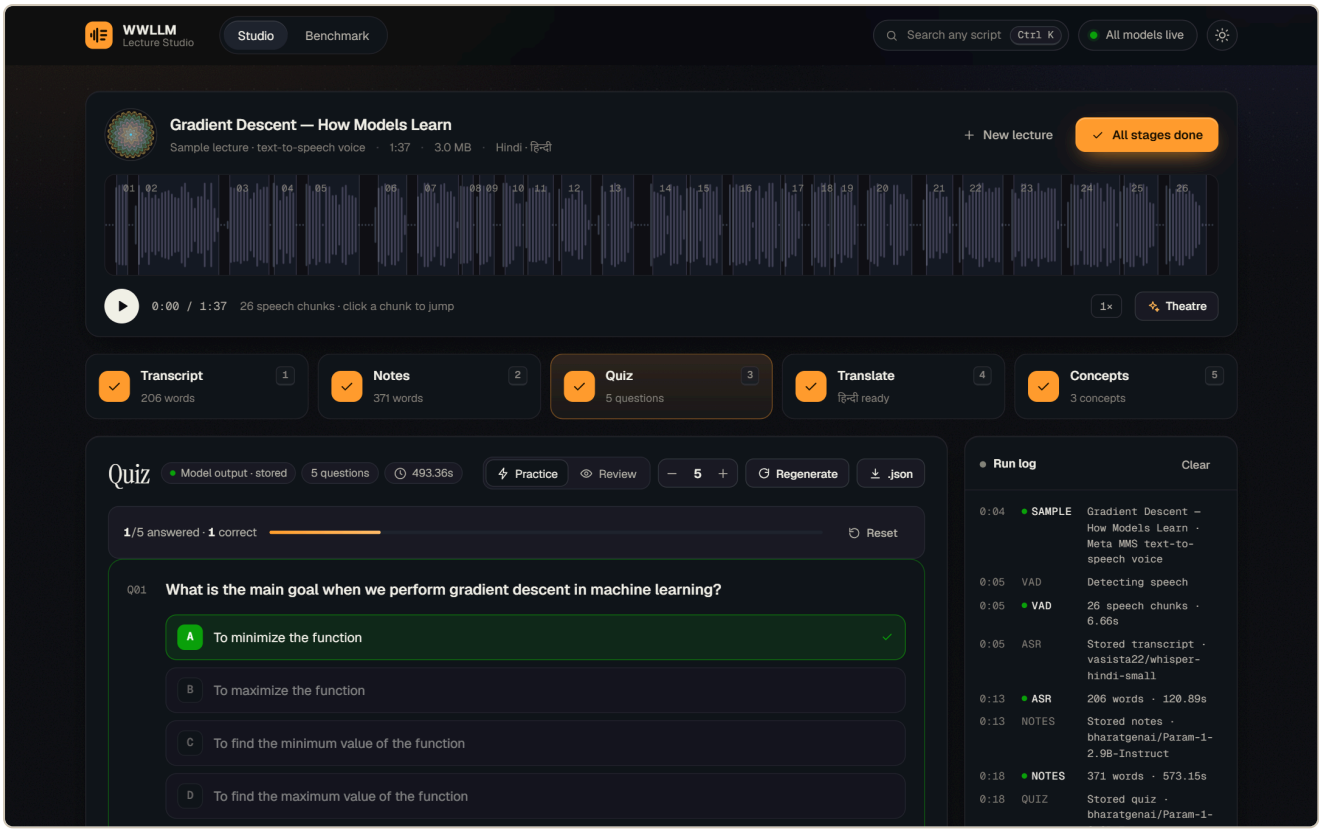


Figure 5. The quiz, written by Param-1. After an answer, the student can replay the moment in the lecture that holds it.

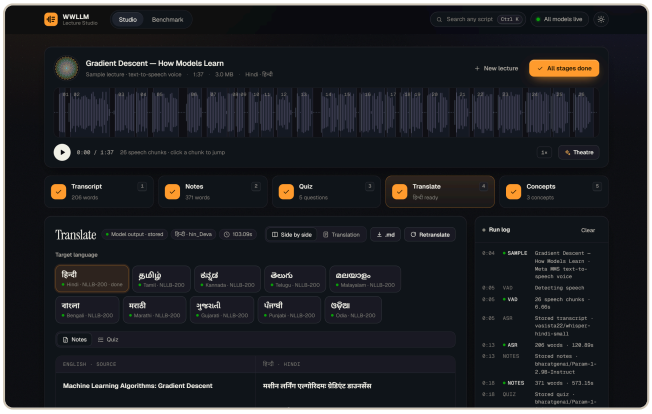


Figure 6. Notes and quiz translated into Hindi by NLLB-200. Headings, lists and the answer key are kept.

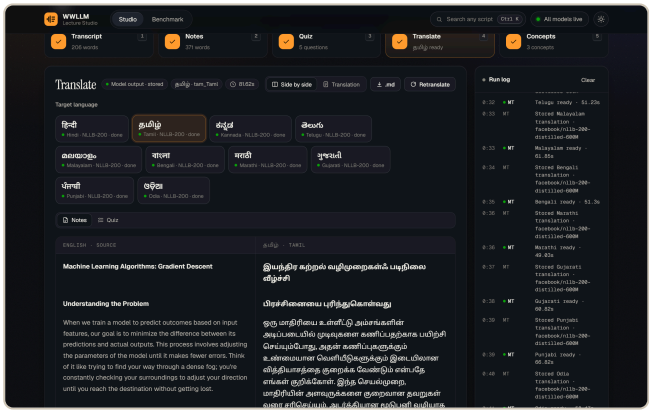


Figure 7. The same notes in Tamil, side by side with the English.

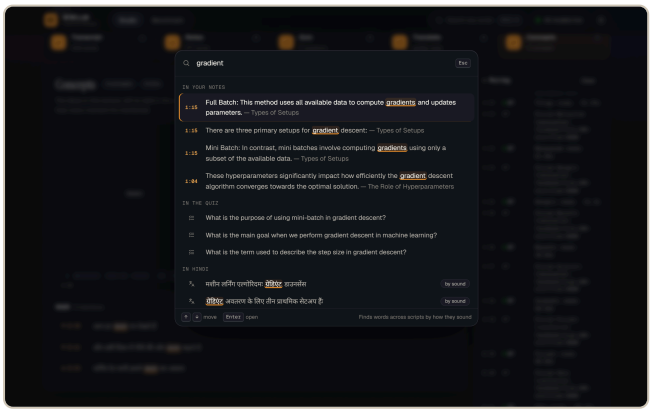


Figure 8. Ctrl K search. The English word "gradient" finds matches in the notes, the quiz and the Hindi translation.

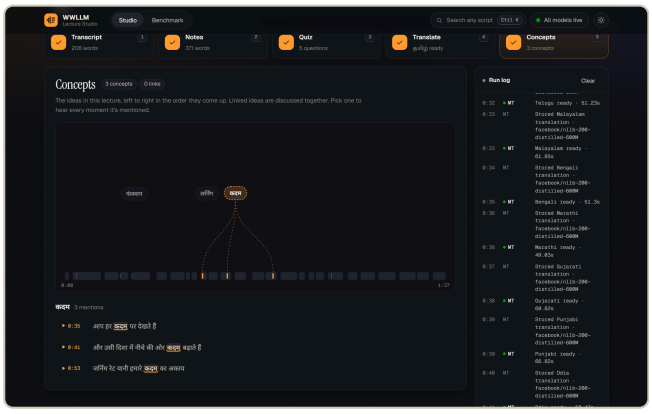


Figure 9. The concept map places each idea on the timeline and lists every moment it is mentioned.

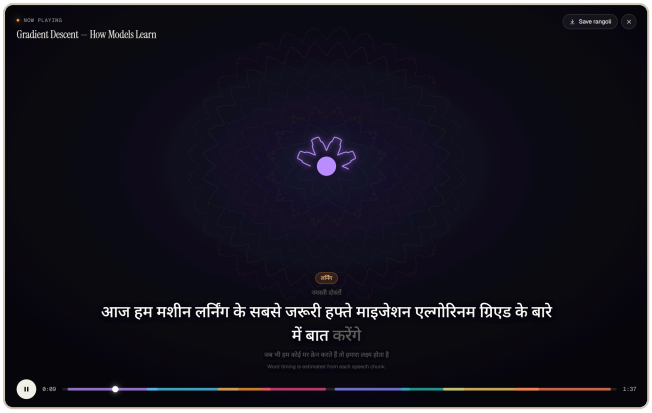


Figure 10. Theatre mode, with captions and a rangoli shaped by the loudness of the speech.

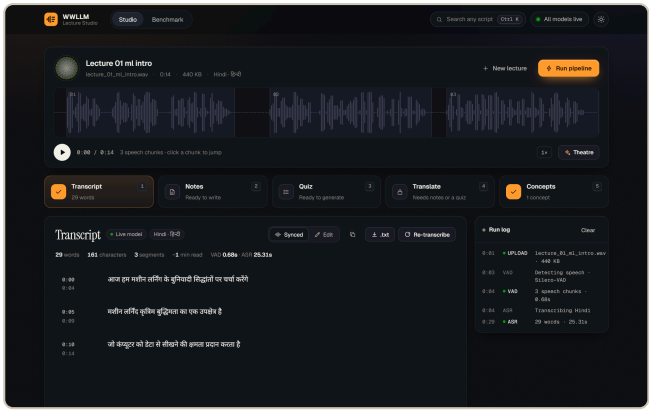


Figure 11. A new recording uploaded and transcribed live by Silero-VAD and Whisper.

6. Evaluation: word error rate

The test set is 40 short technical lectures. Meta's MMS text-to-speech voices read each reference text aloud, the pipeline transcribed the audio, and the output was scored against the text. Word error rate (WER) counts the words changed, dropped or added, as a share of the reference. Punctuation and case are ignored.

Language	Model	Lectures	Average WER	Failures
English	openai/whisper-small	22	0.257	6
Hindi	vasista22/whisper-hindi-small	14	0.277	5
Tamil	vasista22/whisper-tamil-small	2	0.225	0
Kannada	vasista22/whisper-kannada-small	2	0.167	1
All		**40**	**0.258**	**12**

A lecture fails above a WER of 0.30, so 28 pass and 12 fail. Word accuracy is 77.2%: 619 of 802 reference words. This is clean, machine-read speech, so a real classroom with echo and noise is likely to score worse. The Tamil and Kannada rows rest on two clips each, which is too few to compare languages.

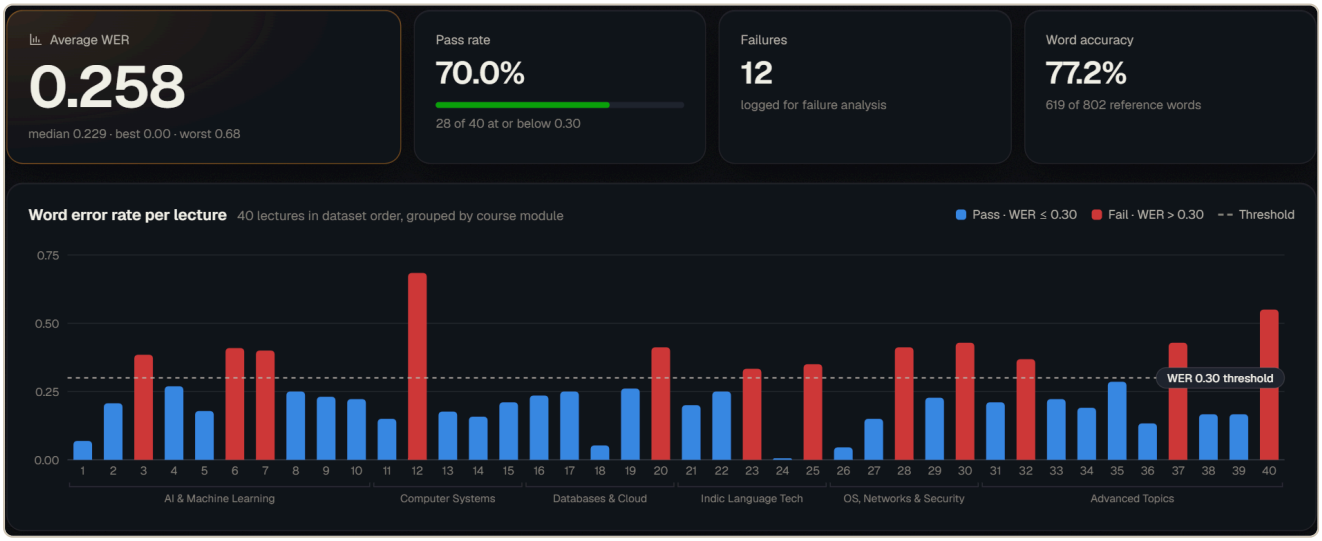


Figure 12. The Benchmark page, built from this measurement.

7. Failure analysis

Of the 207 word errors, 150 are substitutions (72%), 33 are deletions (16%) and 24 are insertions (12%). The worst lecture is Cache Coherence at 0.684, where "cache coherence protocols like MESI ensure" came out as "KakiCarns protocols. Light mess I endure". Going through the 12 failures shows three causes:

Cause	How much it matters	Example
Technical terms and names misheard	Every failing lecture has at least one	"SHA-256" heard as "Cheveth", and "कुबेरनेट्स" as "केनरी"
Opening words lost or garbled	The first word goes wrong in 24 of 40 clips	"बैकप्रोपेगेशन एल्गोरिथम" heard as "प्रोपेशन गोरी"
Compound words spelled differently	Counted as errors though the meaning is right	"pre-image" as "preimage" is 4 of lecture 28's 7 errors

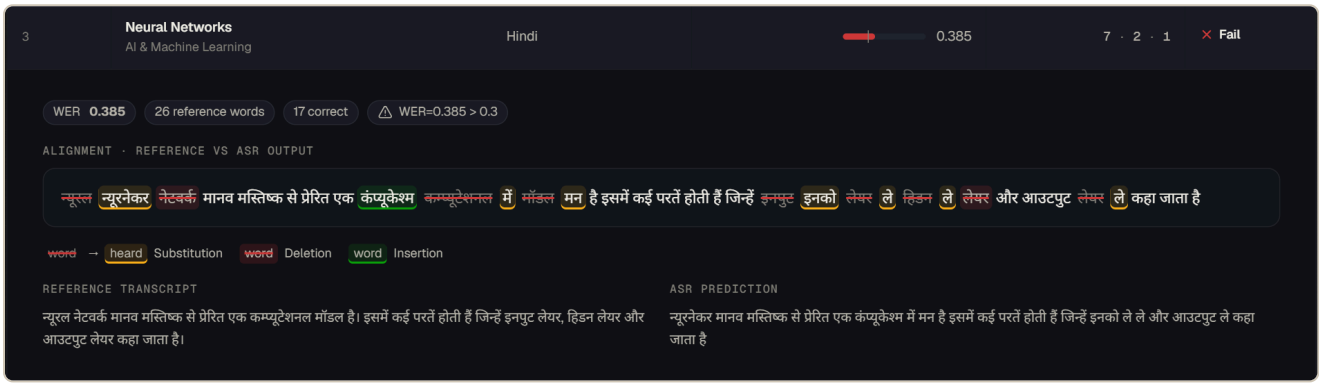


Figure 13. Any lecture opens into a word-by-word comparison of the reference and what the model heard.

One fix was tried for the opening words: half a second of silence added before each chunk, on all 14 Hindi clips. Some opening words came back, but the average WER stayed at 0.277 and one more clip failed, so the change was left out. Two fixes remain untested: a custom vocabulary for technical terms, and scoring compound words as one word.

8. Evaluation: notes and quiz

The notes and quiz for the sample lecture were checked by hand against its reference text. This covers one lecture, so the figures show how the system behaves and aren't a measured rate.

Factual consistency

Statements in the notes	Count	Example
Supported by the lecture	8	A high learning rate overshoots, and a low one is slow
Correct, but added from general knowledge	8	Batch size as a hyperparameter, which the lecture never mentions
Wrong	1	Mini-batches are "less efficient than full batches"

So 16 of 17 statements are correct, working from a transcript with a WER of 0.27. The notes name the topic correctly and keep the lecture's fog analogy. They miss that mini-batch is the most common choice in deep learning, and they drop the lecturer's definition of the gradient as the slope. Notes need a human read before they are trusted.

MCQ quality

Question	On topic	Answer key	Explanation
1. Main goal of gradient descent	Yes	Correct, but a second option says the same thing	None
2. Term for the step size	Yes	Correct: learning rate	None
3. Which is not a type of dataset	Yes	Wrong: confuses datasets with methods	None
4. Where mini-batch is commonly used	Yes	Circular: the question gives away the answer	Present
5. Purpose of mini-batch	Yes	Not supported by the lecture	Contradicts itself

All five questions are on topic, but only question 2 is fully sound, and a teacher should review the quiz before using it. The likely cause is that Param-1 is writing from a noisy Hindi transcript. Generating the quiz from the English notes is the next thing to try.

Translation

Translation is optional in the brief and has no formal score here. NLLB-200 gives readable text but garbles technical terms: "Gradient Descent" became ग्रेडिएंट डाउनसेंस in Hindi, close in sound but not the real term.

9. Testing

122 automated tests pass: 87 in Python, covering the pipelines, service layer, HTTP API and Streamlit app, and 35 in JavaScript, covering search, anchors and concepts. Without the ML stack installed, the Python suite still passes, with 16 model tests skipped. A fresh clone of the GitHub branch passes the same tests.

A scripted run in Chrome went through the sample lecture, all ten translations, search, Theatre, the Benchmark page and a live upload. All 21 checks passed, with no console or network errors.

10. Problems found and fixed

Running real models on real speech exposed these faults.

Symptom	Cause	Fix
Models used one CPU core	Importing silero_vad pins the whole process to one thread	Restore the thread count after the import
Param-1 crashed on load, then wrote gibberish	Its custom code predates transformers 5, which left the rotary position tables uninitialised	Reset the config and rebuild the tables after loading
Each new word re-read the whole prompt	generate() can't use the model's cache format	A small sampling loop that carries the cache itself
Whisper kept 80 words of a 206-word lecture	The fine-tuned model stops at the first pause, and chunks held several sentences	One chunk per stretch of speech, which took the sample's WER from 0.77 to 0.27
Notes had no contents list or section chips	Param-1 wrote #### headings, and the interface looked for ##	Level the heading depth on display
The quiz failed three ways	Param-1 wrote broken JSON, copied the example, then stopped after one question	A plain text format, with each question started for the model
The benchmark was simulated	Hand-written predictions scored against sine-wave audio	Real speech for all 40 clips and a real measurement

11. Repository and running it

Repository: github.com/Tamishsri/WWLLM-CIE-3
Full project: github.com/Tamishsri/WWLLM-CIE-3/tree/feat/lecture-studio

```
pip install -r requirements.txt
python server.py --open
```

Open `http://127.0.0.1:8000` and press Try the sample lecture. Models download on first use, about 8 GB in all. With `requirements-web.txt` alone, the studio still runs the sample from stored results. The README explains how to switch to IndicConformer and IndicTrans2 with a Hugging Face login.

12. Limits

- IndicConformer and IndicTrans2 have not been run or measured.
- The accuracy figures come from synthetic speech, so they show the pipeline working and don't predict classroom accuracy.
- Factual consistency and MCQ quality were checked by hand on one lecture.
- Param-1 is slow without a GPU, at about 9.6 min for one set of notes. The sample plays instantly from stored results, but a new lecture takes this long.
- The user picks the spoken language. There is no automatic detection, and the wrong choice produces nonsense.
- Word timing is estimated from chunk timing, so caption highlighting is close but not exact.
- MMS voices and NLLB-200 are CC-BY-NC, which rules out commercial use.